

Bhavana
Cloud Engineer
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PROFESSIONAL SUMMARY:

- **Cloud Engineer** with **5 years** of experience in **AWS, Python, Java, and SQL**, specializing in cloud infrastructure automation, distributed systems, and backend cloud-native applications.
- Expert in designing and deploying scalable and secure cloud solutions using **AWS** services including **EC2, S3, Lambda, RDS, DynamoDB, IAM, VPC, and CloudFormation**.
- Developed resilient microservices architectures with **Java (Spring Boot), Node.js, and Python**, deployed across containerized environments using **Docker and Kubernetes**.
- Led infrastructure automation using **CloudFormation** and **Terraform** to provision and manage **AWS** resources across multiple environments, ensuring consistency and scalability.
- Designed multi-region architectures using **AWS Transit Gateway, VPC peering, Route 53, and WAF**, ensuring high availability, disaster recovery, and network security.
- Hands-on experience supporting **Kubernetes** workloads in production, including implementing observability with **Prometheus/Grafana**, handling on-call rotations, and applying SRE best practices for incident management.
- Integrated security and compliance checks into CI/CD workflows using **AWS Config rules**, Aqua Security scans, and custom GitHub Actions.
- Collaborated with SREs and application teams to establish **CI/CD** pipelines using **AWS Code Pipeline, Code Build, and GitHub Actions**, streamlining cloud deployments, and improving delivery velocity.
- Provisioned and optimized cloud databases including **Amazon RDS, DynamoDB, and Redshift** to support high throughput, low-latency data processing in cloud applications.
- Designed reusable **Terraform modules** and CDK stacks to ensure consistent provisioning of **AWS** resources across development, staging, and production.
- Experience with monitoring and logging tools such as **CloudWatch, ELK Stack, and AWS Config** for auditing, alerting, and compliance tracking in cloud-native environments.
- Integrated **RESTful APIs** with backend services hosted on **AWS Lambda and EC2**, supporting event-driven architectures with **SQS, SNS, and API Gateway**.
- Architected and deployed **DevSecOps pipelines** integrating security-as-code principles across **Windows, Linux, and cloud-native platforms**, ensuring **compliance with enterprise vulnerability management** and data privacy policies.
- Utilized **Git-based** version control (**GitHub, Bitbucket**) for infrastructure and application codebases, incorporating branching strategies and code reviews for reliability.
- Automated multi-account **AWS** infrastructure provisioning using **Terraform and AWS CDK**, enabling secure and scalable deployments aligned with organizational guardrails.
- Enabled enterprise-wide adoption of **GitOps principles** through implementation of **ArgoCD and FluxCD** in regulated environments.
- Implemented **network protection** and **data loss prevention (DLP)** solutions across hybrid environments, ensuring adherence to industry standards such as **HIPAA, GDPR, and NIST**.
- Designed and operated **high-availability EKS clusters** with autoscaling, RBAC policies, and service mesh integration using Istio.
- Engineered fault-tolerant and auto-healing systems using **Auto Scaling Groups, Elastic Load Balancers (ALB/NLB), and AWS CloudWatch Alarms** to maintain uptime and reduce manual intervention.
- Built containerized applications and deployed **via EKS and ECS**, supporting autoscaling, health checks, and blue green deployments for zero-downtime rollouts.
- Implemented **Terraform remote state management** with **S3 and DynamoDB** locking to prevent resource conflicts and maintain infrastructure integrity.

TECHNICAL SKILLS:

Programming Languages:	Java, Python, PySpark, Shell, Perl
Cloud Providers:	AWS, Azure, Terraform
Web Technologies:	HTML, JavaScript, jQuery, Bootstrap, AEM, Angular, React JS, AngularJS,
Database Management:	MySQL, PostgreSQL, IBM DB2, Oracle Database, MongoDB, SQL Queries
Cloud Services:	AWS (Cloud Formation, Lambda, S3, IAM, EC2, RDS, SNS, SQS), Azure
Configuring Management Tools:	Chef, Puppet, Ansible
Build And Deployment Tools:	Apache Ant, Maven, Gradle, Tomcat, Apache Kafka, RabbitMQ, Zookeeper
Databases:	MySQL, Oracle Database, MongoDB
Logging Tools:	Elastic Search, Kibana
Continuous Integration/Deployment:	AWS Code Pipeline, Cloud-bees Jenkins
Repository Management:	Git, GitHub, GitLab, Bitbucket, AWS Code Commit
Containerization/Orchestration:	Docker, OpenShift, Kubernetes, Apache, ECS
Operating Systems:	Linux (Red Hat Enterprise Linux, CentOS, Ubuntu) Unix, macOS
Monitoring Tools:	AWS CloudWatch, Splunk, Prometheus, Grafana, AppDynamics

PROFESSIONAL EXPERIENCE:

Amazon - Austin, Texas | Feb 2022 – Mar 2025

Cloud Engineer

Responsibilities:

- Architected and implemented **high-throughput, low-latency distributed systems** using Java and AWS, leveraging multi-threading and parallel processing to ensure optimal scalability and performance in cloud environments.
- Designed and implemented **fanout, direct, topic, and headers exchanges** in RabbitMQ to support varied message routing needs across microservices.
- Built and optimized **RESTful and GraphQL APIs** using Spring Boot and WebFlux (Reactive Programming), ensuring fast response times and seamless integration within cloud-native applications.
- Designed and deployed **AWS Glue ETL pipelines** process large-scale structured and semi-structured data from S3, RDS, and external APIs for enterprise data lake initiatives.
- Configured and managed AWS services including **EKS, EC2, RDS, IAM, Lambda, and CloudWatch** to build resilient, cost-effective, and scalable cloud infrastructure.
- Automated infrastructure provisioning using **Terraform and Helm**, deploying and managing Kubernetes clusters, namespaces, RBAC, and custom controllers at scale.
- Orchestrated **serverless data workflows** using AWS Glue, Step Functions, and Lambda to automate and optimize ETL pipelines with minimal manual intervention.
- Led **end-to-end AWS cloud migrations** across multi-account environments with service control policies (SCPs), ensuring secure networking, data integrity, and minimal downtime.
- Participated in **on-call rotations**, resolving production issues involving config drift, availability outages, and deployment failures using monitoring and rollback strategies.
- Implemented **Kubernetes admission controllers and network policies** to harden workloads and limit service-to-service communication as per zero-trust principles.
- Configured **publisher confirms** and **consumer acknowledgments** to guarantee message durability and safe delivery in critical workflows.
- Led knowledge-sharing sessions on **IaC best practices**, Kubernetes security, and streaming architectures, improving team onboarding and code quality.

- Applied **AWS Well-Architected Framework** to design secure, high-performing, fault-tolerant cloud systems aligned with business goals.
- Led the design and governance of **secure-by-design frameworks** across Windows and Linux workloads, enabling consistent security enforcement in both containerized and traditional deployments.
- Developed a Slack-integrated alert system using **Lambda, SNS, and KSQL**, surfacing production incidents in under 30 seconds.
- Managed secrets using **Kubernetes Secrets and AWS Parameter Store**, enforcing encryption-at-rest and least privilege access across environments.
- Built custom **Jenkins pipelines** with shared libraries for modular and reusable CI/CD workflows, standardizing deployment logic across multiple microservices and reducing maintenance overhead.
- Developed **modular Helm charts** and automated Kubernetes manifest deployment through CI/CD pipelines, improving release velocity and standardization.
- Conducted performance tuning of **JVM containers** on Kubernetes, optimizing GC settings, heap size, and resource limits for efficient pod operation.
- Automated queue and exchange provisioning using **Ansible and RabbitMQ Management API**, supporting zero touch deployments across dev and staging environments.
- Led infrastructure-as-code adoption across teams by building GitOps-ready pipelines with **Terraform workspaces, Helm, and ArgoCD**.
- Built serverless DR pipelines using **AWS Step Functions** and S3 cross-region replication, ensuring RPO/RTO targets are met.
- Partnered with application teams to refactor monoliths into microservices, aligning infrastructure with agile product delivery.
- **Provisioned cloud infrastructure using Terraform and Helm**, enabling consistent, version-controlled deployments of Kubernetes clusters, application configs, and platform services in alignment with GitOps principles.
- Developed **shared CI/CD libraries in Jenkins** for multi-service deployments, standardizing pipeline logic across microservices.
- Played a pivotal role in cloud migration efforts, orchestrating movement of 50+ workloads from on-prem to AWS with near-zero downtime.
- Configured and secured **AWS Lambda functions** using VPC endpoints, encrypted environment variables, and restricted IAM roles.
- Built a fault-tolerant alerting system that consumes logs from Kafka topics, applies stream processing via KSQL, and routes critical events to Slack/Email using AWS Lambda and SNS.
- **Configured and managed Kubernetes admission controllers, network policies, and secrets**, enhancing runtime security, restricting lateral movement, and ensuring encrypted configuration management across environments.
- Implemented **shovel and federation plugins** in RabbitMQ to replicate messages across different brokers for regional high availability and fault tolerance.
- Built custom alerts and SLO dashboards for RabbitMQ, Kafka, and microservices, enabling early anomaly detection and prevention.
- Conducted cost optimization of cloud workloads using **CloudWatch metrics, Reserved Instances, and rightsizing strategies**
- **Automated environment provisioning for testing and staging** using Docker Compose and Kubernetes manifests, enabling rapid spin-up of isolated, ephemeral test environments during pull request validations.

Environment: Java, Python, TypeScript, SQL, AWS (EC2, S3, Lambda, RDS, DynamoDB, API Gateway, Cloud Formation, Code Pipeline, IAM, Kinesis, SNS, SQS, Step Functions, CloudWatch), AWS Glue, ETL Pipelines, Spring Boot, Node.js, AWS Code Build, GitHub Actions, JUnit, TestNG, Mockito, IntelliJ, Apache Kafka, RabbitMQ, Kafka Streams, Kafka Connect, Dead Letter Queues (DLQ), Event-Driven Architecture, Message Queuing Patterns.

OCC (The Options Clearing Corporation) - Dallas, Texas | May 2021 – Jan 2022

Cloud Engineer

Responsibilities:

- Designed and managed highly available **Amazon EKS clusters**, implementing **scalability, security best practices, and RBAC policies** for enterprise workloads.
- Performed advanced **RabbitMQ cluster management** including mirrored queues, disk nodes, and quorum queues for message durability and partition resilience.
- Automated multi-account **AWS infrastructure** provisioning using **Terraform, Ansible, and AWS Organizations**, ensuring repeatability and compliance.
- Optimized Kubernetes networking and service mesh using **AWS VPC CNI, Core DNS, and Istio**, ensuring secure and high-performance service communication.
- Conducted peer reviews of code and test artifacts to ensure adherence to quality standards, coding best practices, and maintainability.
- Set up **multi-cluster Kafka replication** using **MirrorMaker 2** to support cross-region disaster recovery and global message availability.
- Developed event-driven workflows using **AWS Step Functions** and Lambda (Python) to automate infrastructure health checks and remediation actions.
- Integrated **REST API calls** within ETL pipelines to ingest external data sources and enrich business datasets in realtime, enabling more actionable analytics.
- Tuned **Kubernetes networking performance** using AWS VPC CNI, CoreDNS tuning, and resource quotas for optimal pod scheduling.
- Wrote comprehensive **unit and integration tests** using **Jest, React Testing Library**, and Cypress, achieving >90% test coverage.
- Spearheaded cloud governance automation using **Terraform Sentinel and custom pre-commit hooks**, enforcing cost controls and security policies during code review.
- Created **cross-region VPC peering and Transit Gateway** architectures to facilitate secure, scalable network connectivity across business units.
- Worked in an Agile environment, contributing to PI planning, retrospectives, sprint demos, and backlog refinement sessions to align with cross-team priorities.
- Automated Kubernetes resource cleanup (unused ConfigMaps, PVCs, Jobs) using scheduled **Python cron jobs** to maintain cluster hygiene and optimize storage costs.
- Designed and implemented event-driven architectures using **AWS Lambda, SQS, SNS, and Step Functions**, improving process automation and system decoupling.
- Engineered **Kafka producers/consumers** with schema validation, retries, and high-throughput streaming pipelines using Apache Flink.
- Integrated **Adobe Analytics, Target, and Launch** into AEM pages to enable advanced personalization and customer behavior tracking.
- Tuned Kafka performance by optimizing **broker configurations** (e.g., batch size, linger.ms, replication factor) and **Zookeeper ensembles** to reduce latency and improve throughput.
- Integrated **Datadog APM and metrics into Kubernetes workloads**, enabling real-time visibility and SLA monitoring across services.

Environment: Python, EKS, Lambda, SQS, SNS, S3, AEM, Transit Gateway, WAF, Security Groups, IAM, AWS STS, AWS ECR, EKS, OpenShift, Helm, GitOps (ArgoCD, FluxCD), Kubernetes HPA, Cluster Autoscaler, Karpenter, Istio, CoreDNS, AWS VPC CNI, Terraform, Ansible, Terraform workspaces, Jenkins, GitHub Actions, Docker, Harbor, AWS ECR, Aqua Security, Clair, CIS Docker Benchmark, Python, Node.js, Java, Bash, AWS WAF, IAM policies.

Ekam It Solutions - Hyderabad, India | Jan 2019 -Dec 2019

Software Engineer

Responsibilities:

- Monitored and troubleshooted cloud resources, including **EC2 instances**, databases, and networking components, to ensure system reliability and performance.
- Assisted in developing backend services using Java, contributing to API design and service integration within microservice-based architectures.
- Managed secrets securely via **Kubernetes Secrets, AWS Secrets Manager, and SSM Parameter Store**, ensuring encryption at rest and access auditing.
- Collaborated with senior engineers to design and document internal APIs using **Swagger/OpenAPI**, ensuring clear versioning and developer onboarding.
- Worked with DevOps tools such as Git, Jenkins, or Docker to build, test, and deploy software reliably.
- Implemented basic producer-consumer flows using **RabbitMQ**, handling message queues for order processing and retry logic for transient failures.
- Set up and tested AWS **SNS topics and SQS queues** for event-driven workflows in dev environments, supporting asynchronous notification pipelines.
- Helped to containerize applications using **Docker** and deploy them on **Kubernetes clusters** (EKS, AKS, or GKE) under the guidance of senior engineers.
- Built secure network topologies using **VPC peering, Transit Gateway, NACLs, and AWS WAF** to prevent lateral movement and secure data ingress/egress.
- Developed **custom workflows** for content approval, translation, and publishing using AEM Workflow Launcher and Process Steps.
- Assisted in **optimizing backend performance** by profiling Python applications and tuning database queries.
- Integrated backend services with front-end components, **REST APIs**, and third-party tools as part of full-stack features.
- Participated in **AEM upgrades**, resolving deprecated API usage and ensuring compatibility with new features.
- Assisted in migrating components from class based to functional React components, enhancing maintainability and readability.
- Ensured cloud security best practices, including configuring firewalls, implementing encryption, and managing **IAM** policies to protect sensitive data.
- Participated in monitoring, logging, and incident response processes to ensure system reliability and uptime.

Environment: AWS, Azure, Google Cloud, Terraform, CloudFormation, Jenkins, GitHub Actions, Kubernetes, VPC, Subnets, Azure Backup, Python, Bash, PowerShell, AWS CloudWatch, Azure Monitor, DynamoDB, SQL Database.

CERTIFICATIONS:

- AWS Certified Developer – Associate | Amazon Web Services (AWS) | Sept 2022
- AWS Certified Solutions Architect – Associate | Amazon Web Services (AWS) | April 2024

EDUCATION:

- Master of Science in Data Science - University of North Texas, Denton | Dec 2021| GPA: 3.8/4.0
- Bachelor of Technology in Computer Science and Engineering - Jawaharlal Nehru Technological University, India | 2019 | GPA: 8.0/10.0.